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Activity 1

S. Mahalakshmi
192524372

2) Stack + Queue operations

Stack (starts empty):

Operation	Stack (bottom → top)	Popped Value
Push(54)	54	
Push(52)	54, 52	-
Pop()	54	52
Push(55)	54, 55	-
Push(62)	54, 55, 62	
S = Pop()	54, 55	S = 62

Queue (starts empty):

Operation	Queue (front → rear)	Dequeued value
enqueue(21)	21	-
enqueue(24)	21, 24	-
dequeue()	24	21
enqueue(28)	24, 28	-
enqueue(32)	24, 28, 32	-
q = dequeue()	28, 32	q = 24

$$S + q = 62 + 24 = 86$$

Postfix Evaluation:- $10\ 5 + 60\ 6 / * 8 -$

Token	Action	Stack (bottom $\rightarrow$ top)
10	Push	10
5	Push	10, 5
+	Pop 5, 10 $\rightarrow 10 + 5 = 15$, Push	15
60	Push	15, 60
6	Push	15, 60, 6
/	Pop 6, 60 $\rightarrow 60 / 6 = 10$, Push	15, 10
*	Pop 10, 15 $\rightarrow 15 * 10 = 150$, Push	150
8	Push	150, 8
-	Pop 8, 150 $\rightarrow 150 - 8 = 142$, Push	142

Result = 142

GATE CSE 2007 - $8\ 2\ 3\ 1 / 2\ 3\ * + 5\ 1\ * -$

Token	Action	Stack (bottom $\rightarrow$ top)
8	Push	8
2	Push	8, 2
3	Push	8, 2, 3
/	Pop 3, 2 $\rightarrow 2^3 = 8$, Push	8, 8
1	Pop 8, 8 $\rightarrow 8 / 8 = 1$, Push	1
2	Push	1, 2
3	Push	1, 2, 3
*	Pop 3, 2 $\rightarrow 2 * 3 = 6$, Push	1, 6 (first * evaluated here)

Top two elements after first * : 6 (top), 1 (below)

GATE CSE 2004 - Infix $a + b \times c - d^1 e^1 f$ to

Postfix

Character	Stack	output
a	empty	a
+	+	a
b	+	ab
x	+ x	ab
c	+ x	abc
-	Pop x, Pop + $\rightarrow$ then push -	abc x +
d	-	abc x + d
^	- ^	abc x + d
e	- ^	abc x + d e
^	- ^ ^	abc x + d e
f	- ^ ^	abc x + d e f
end	pop all	abc x + d e f ^ ^ -

Postfix = $abc x + d e f ^ ^ -$

Stack Push / POP Sequence

PUSH(10), PUSH(20), POP, PUSH(10), PUSH(20),
POP, POP, POP, PUSH(20), POP

Solution :-

Operation	Stack (bottom → top)	Popped
PUSH(10)	10	—
PUSH(20)	10, 20	—
POP	10	20
PUSH(10)	10, 10	—
PUSH(20)	10, 10, 20	—
POP	10, 10	20
POP	10	10
POP	empty	10
PUSH(20)	20	—
POP	empty	20

Sequence of popped values: 20, 20, 10, 10, 20

Good
Ans
2/8/20